

NEERAJ KUMAR SINGH

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ABOUT

Deep learning researcher with 4+ years of experience in NLP and Computer Vision, specializing in explainable AI and model interpretability. Strong foundation in ML/DL with proven research track record.

WORK EXPERIENCE

Lexsi AI, Mumbai, India

Research Scientist

April 2024 - March 2026

- Shipped DLBacktrace v2, a production-grade explainability stack enabling reliable relevance tracing and visualization across tabular, vision, NLP, and LLM workloads.
- Expanded model coverage via PyTorch `torch.export`/FX graphs; fixed relevance propagation across transformer attention graphs and common ops, and added robust LLaMA-family support + validation utilities.
- Implemented end-to-end MoE compatibility (JetMoE, OLMoE, Qwen3-MoE, GPT-OSS), including routing-aware relevance propagation and FP4/FP8/FP16 relevance quantization (up to 17–93× compression, MAE < 0.001).
- Built evaluation + pipeline tooling (`dlbacktrace-viz`, ModelRegistry, HF-compatible sampling/beam search) and improved long-sequence efficiency via token-wise delayed relevance, caching, and compressed trace storage (`gzip/lzma/7z`).

CVPR Unit, Indian Statistical Institute, Kolkata, West Bengal

Research Assistant, NLP Lab

September 2021 - February 2024

- Top contributor to “TrustedED: Evaluating Trustworthiness of Deep Learning Systems”; developed explainable hate speech detection system for Indic languages with intensity reduction.
- Built web application integrating tCNNs, Precily, and Paccmann models for IC50 prediction using drug properties.

PyFox Technologies Private Limited, Lucknow, Uttar Pradesh

Data Analysis Intern

June 2020 - August 2020

- Authored Python scripts for customer purchase data analysis and sales data analytics for retail clients.

EDUCATION

Indian Statistical Institute, Kolkata, West Bengal

2019 - 2021

Master of Technology

Percentage: 77.45

Bundelkhand Institute of Engineering & Technology, Jhansi, Uttar Pradesh

2015 - 2019

Bachelor of Technology

Percentage: 71.82

PUBLICATIONS

- Interpretability-Aware Pruning for Efficient Medical Image Analysis [Accepted]
 - Proposed an interpretability-guided neuron-pruning framework for medical image models that ranks and removes low-importance neurons using attribution methods (LRP, DL-Backtrace, Integrated Gradients), aggregating relevance to stable per-neuron importance scores with representative sampling (e.g., 10 samples/class via random, confidence, and clustering).

- Evaluated across four medical imaging datasets (MURA, KVASIR, CPN, Fetal Planes) and multiple architectures (VGG19, ResNet50, ViT), achieving high compression (often ~65–80% pruning; up to ~85% for ViTs) with minimal accuracy loss; LRP and clustering-based sampling provided the most robust pruning behavior.
- DLBacktrace: A Model Agnostic Explainability for any Deep Learning Models [Accepted]
 - Introduced DLBacktrace, a novel interpretability technique compatible with PyTorch and TensorFlow, designed to improve transparency across MLPs, CNNs, LLMs, and custom DNNs; benchmarked it against SHAP, LIME, Grad-CAM, and Integrated Gradients, demonstrating superior adaptability and performance across NLP, vision, and tabular data tasks.
- XAI-Evals: A Framework for Evaluating Post-Hoc Local Explanation Methods [arXiv Pre-Print]
 - Developed `xai_evals`, a Python package that unifies generation, benchmarking, and evaluation of XAI methods (e.g., SHAP, LIME, Grad-CAM, IG, Backtrace) across tabular and image data, supporting metrics like faithfulness, sensitivity, and robustness to enhance model interpretability and trust.
- Convolutional Neural Networks Exploiting Attributes of Biological Neurons (Under Review) [arXiv Pre-Print]
 - Proposed a two-tower CNN architecture combining a shallow network based on the Push-Pull Combination of Receptive Fields (PP-CORF) model of simple cells with ResNet-18, achieving 5% – 13% average performance improvement over ResNet-18 on CIFAR-10, CIFAR-100, and ImageNet-100 datasets.
- SafeSpeech: A Three-Module Pipeline for Hate Intensity Mitigation of Social Media Texts in Indic Languages [Accepted]
 - Proposed a cost-effective three-module pipeline for hate speech mitigation that detects hateful content, quantifies word-level hate intensity, and reduces overall hatefulness; validated through extensive experiments on five Indic languages (Hindi, Marathi, Tamil, Telugu, and Bengali), achieving high BERTScores (0.96 – 0.99) in both automated and human evaluations.
- Neeraj Kumar Singh, Utpal Garain: An Analysis of Transformer-based Models for Code-mixed Conversational Hate-speech Identification [FIRE (Working Notes) 2022: 513-521]

PROJECTS

- **XAI-RAG: Explainable Retrieval-Augmented Generation** *Dec 2025 - March 2026*
 Production RAG pipeline with built-in explainability: hybrid search (ChromaDB + Elasticsearch + Reciprocal Rank Fusion), cross-encoder re-ranking, retrieval attribution (per-chunk score breakdown across all search stages), and NLI-based faithfulness verification (DeBERTa) that checks each generated claim against retrieved evidence. Includes RAGAS auto-evaluation, OpenTelemetry tracing, and async FastAPI backend with SSE streaming.
 Github Link- <https://github.com/neeraj1909/xai-rag>

PROGRAMMING LANGUAGES AND TOOLS

- Python, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Scikit-Learn, Hugging Face Transformers, LangChain, LlamaIndex, OpenCV, spaCy, NLTK
- Explainability: DLBacktrace, SHAP, LIME, Grad-CAM, Integrated Gradients, Layer-wise Relevance Propagation
- LLM/Transformer Architectures: BERT, LLaMA, Qwen, T5, Mixture-of-Experts (JetMoE, OLMoE, Qwen3-MoE, GPT-Oss)
- Tools: Git, GitHub, Jupyter, VS Code, PyCharm, Plotly